

Properties of Big O

lemma 6.2 Asymptotic equivalence of max and sum.

$$f(n) = O(g(n) + h(n)) \iff f(n) = O(\max(h(n), g(n)))$$

Ex: $f(n) = n^2 + n = O(n^2 + n) \iff$

$$\begin{aligned} f(n) &= O(\max(n^2, n)) \\ &= O(n^2) \end{aligned}$$

Lemma 6.3 Transitivity of $O(\cdot)$

If $f(n) = O(g(n))$ and $g(n) = O(h(n))$,

then

$$f(n) = O(h(n))$$

In other words

$$[f(n) = O(g(n))] \wedge [g(n) = O(h(n))] \Rightarrow [f(n) = O(h(n))]$$

Ex: $f(n) = n^2 + 4n$

$$f(n) = O(n^2)$$

$\uparrow$
 $g(n)$

$$g(n) = O(n^3)$$

$$f(n) = O(n^3)$$

Lemma 6.4

Addition and multiplication
preserve $O(\cdot)$ -ness.

$$[f(n) = O(h_1(n))] \wedge [g(n) = O(h_2(n))]$$

$\Rightarrow$

$$f(n) + g(n) = O(h_1(n) + h_2(n))$$

and

$$f(n) \cdot g(n) = O(h_1(n) \cdot h_2(n))$$

$$\text{Ex: } f(n) = n^2 + 1 \quad g(n) = n^4 + 2$$

$$h_1(n) = n^3 \quad h_2(n) = n^4$$

$$f(n) = O(h_1(n)) = O(n^3)$$

$$g(n) = O(h_2(n)) = O(n^4)$$

$$f(n) + g(n) = n^2 + 1 + n^4 + 2 = n^4 + n^2 + 3$$

$$f(n) + g(n) = O(h_1(n) + h_2(n))$$

$$= O(n^3 + n^4)$$

$$= O(n^4)$$

$$f(n) \cdot g(n) = (n^2 + 1)(n^4 + 2)$$

$$= O(h_1(n) \cdot h_2(n))$$

$$= O(n^3 \cdot n^4)$$

$$= O(n^7)$$

Lemma 6.5

$$\text{Let } f(n) = \sum_{i=0}^k a_i n^i = a_0 n^0 + a_1 n^1 + a_2 n^2 + \dots + a_k n^k$$

be a deg- k polynomial, then

$$f(n) = O(n^k)$$